Chapter 5 Matter and changes in its states

The World is made up of many tiny materials. As you have studied that all the **solid, liquid, gaseous** materials are **matter;** and **matter** has **mass** and occupies **space.** Have you ever thought what the solids, liquids and gases are made of? Imagine you could see inside a glass of water, an ice cube or air inside a cup. What would you see? You would see many particles arranged in different ways and moving in different ways.

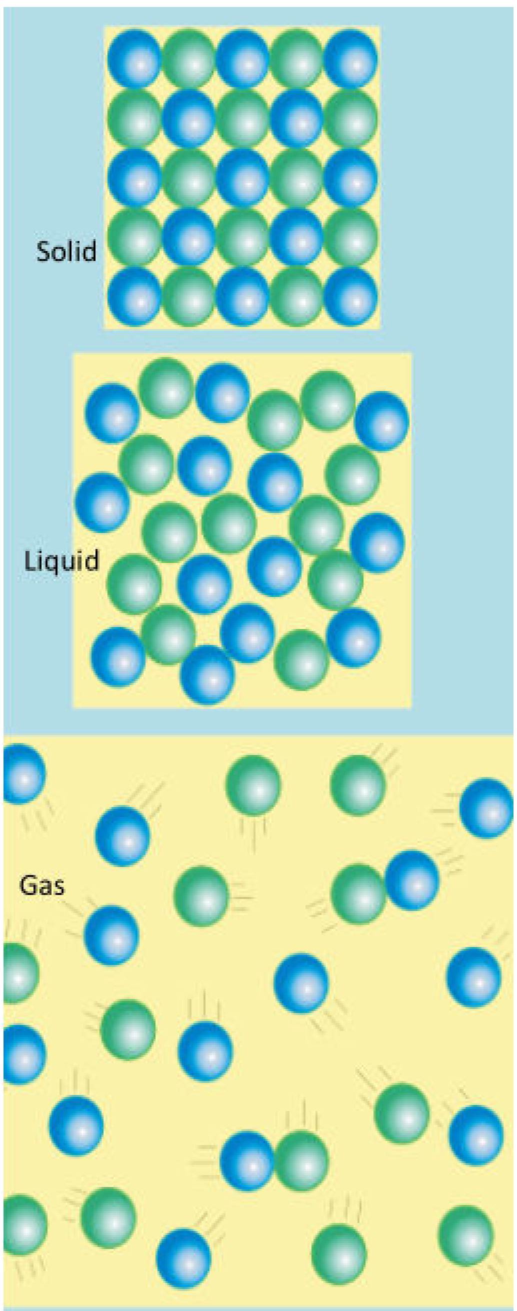


Figure 5.1: Arrangements of particles in three states of matter

In this chapter, you will learn about:

- Matter.
- Arrangements of particles in solids, liquids and gases.
- Effect of heat on arrangement of particles.
- Processes involving change of states (melting, freezing, boiling, evaporation and condensation).
- Application of condensation and evaporation in nature (water cycle).

All the students will be able to:

- ✓ Describe the properties of three states of matter on the basis of arrangement of particles.
- ✓ Demonstrate the arrangements of particles in the three states of matter through models.
- ✓ Investigate the effect of heat on particle motion during a change of states.
- ✓ Demonstrate and explain the processes that are involved in change in states.
- ✓ Describe the role of evaporation and condensation in the water cycle.
- ✓ Identify and describe forms of moisture in the environment (e.g.,dew, snow, fog, frost, rain).

Matter

Activity 1: Observing and describing the properties of three states of matter.

Describe the properties of three states of matter on the basis of arrangement and movement of particles.

You have observed many solid materials in your previous class that are hard and you cannot change the shape. You have also observed many liquid materials that you can pour. Likewise you have observed many liquid materials that can change shape but not volume. Similarly you have observed materials that can change both shape and volume. Write the name or draw one material from each state in the below given below.

It flows and it has a fixed volume and no fixed shape.	It is hard and it has a fixed Shape and Volume.
It flows and it has no fixed shape and volume.	

These **Solid, liquid and gaseous materials** are made up of many tiny **like particles** the **atoms** and **molecules.** The particular **arrangement and movement** of these **particles** in the solid, liquid and gaseous materials provide the material its particular properties.

Activity 2: Movement of ink in a glass of water

What I need:

- A transparent glass.
- Water
- Ink
- Dropper.

What to do:

- Now take some water in a glass or cup and observe it closely.
- Is the water made up of tiny discrete particles, or is it continuous?
- Pour a drop of ink or colour with a dropper and observe closely.

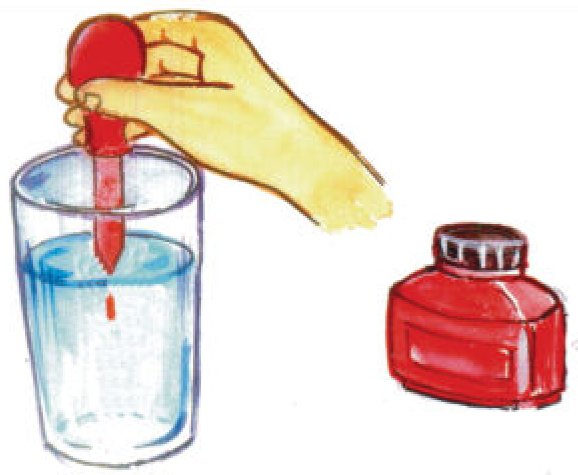


Figure 5.2: Dropping ink in a glass of water.

Activity questions:

- What did you observe?
- Why is the colour spreading/moving in the glass of water?
- How is the colour moving? What is pushing it here and there?

What I concluded: Discuss with class fellows and write here.

Teacher Note: As the students find it difficult to imagine that liquid is made up of tiny particles that can move around, rotate and slide past each other, demonstrate the activity and ask students to observe the movement. Why is ink moving about? What is hitting the ink particles and making it move in zig zag manner?

Activity 3: Demonstrating the arrangements of particles in three states of matter.

Demonstrate the arrangements of particles in the three states of matter through models.

- Observe the models a, b and c.
- Talk to your classfellows.

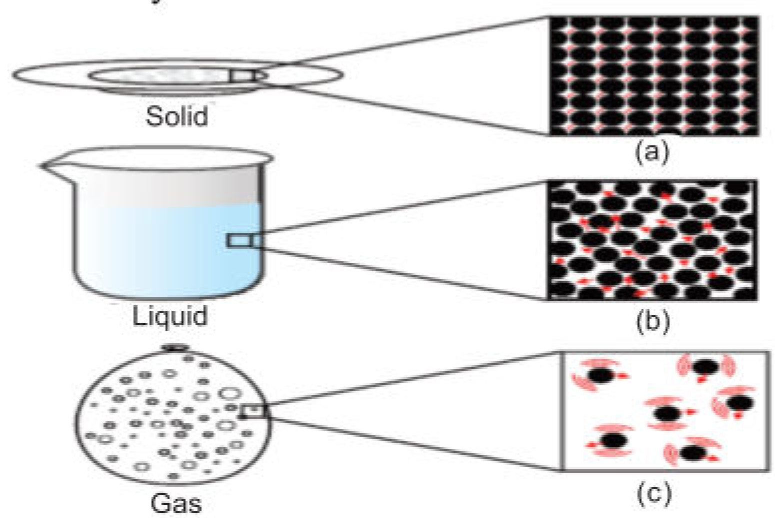


Figure 5.3: Arrangement of Particles in Solid, Liquid and Gases

Write and draw your observation below.

States of Matter Write the States below:	Arrangement of Particles Draw the arrangements below:
Model (a)	
Model (b)	
Model (c)	

The arrangement of particles in solids, liquids and gaseous materials are different. This arrangement and movement of the particle, give the materials its particular properties and also the states. Let us study these movements through a model.

Teacher Note: Ask the students to observe the particle arrangement and movement and talk to classfellows. Then ask them to draw the arrangement of particles in the above boxes.

Activity 4: Building models of solid, liquid and gases.

What I need:

- Elastic band
- String
- Cardboard
- Balloon
- Jar
- Beads/Seeds

What to do:

1) Cut a circle of cardboard and place it on the balloon.
2) Thread a string through the balloon and card.
3) Knot the string so that it won't pull through the card.
4) Tie knots at the other end of the string, about a centimeter apart.
5) Put the Beads/seeds in the jar to form at least two layers.
6) Stretch a balloon over the neck of the jar and keep it in place with the help of the elastic band as shown in the picture.

What I observed:

Now hold the jar upside down and run your fingers gently down the string. What did you observe? Did you observe the movement of the beads/seeds on the top layer? Now pull down the string and let go; observe the beads. Repeat this and observe the movement of the beads.

Teacher Note: For model making the students need to be provided materials in groups. Help the students in making the model, observing and recording observations.

What I concluded:

Matter is made up of particles that are atoms and molecules. These atoms and molecules form different forms of matter that exists in different states as solids, liquids and gaseous states. Do you agree that the arrangement of atoms and molecules in these states are arranged differently? Study the differences and talk to your friend about the differences.

Particles in solids are closely packed together in a regular pattern with very small spaces between them, as you can observe in the picture below:

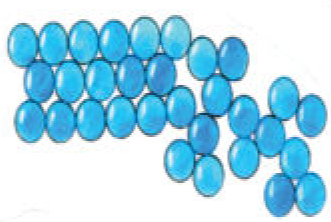
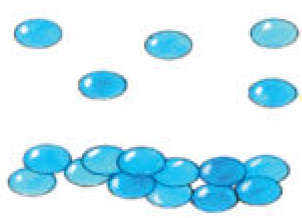


Figure 5.4: Particles changing from solid to liquid

Particles in liquids are a little less closely packed than in solids, they are in loose clusters; arrangement is not regular and have very small empty spaces between them. The picture above shows the change of solid to liquid state.



Particles in gases are widely spread out compared to particles in liquids; theyhave no regular pattern and have large empty spaces between them, as shown in top half of the picture on the left.

Figure 5.5: Particles changing from liquid to gas

Teacher Note: Explain to the students with example that matter is made up of atoms and molecules and it exists in three states. With the help of examples, to demonstrate the arrangement of particles . Make a list of things made up of atoms and molecules. Make models to demonstrate the difference between atom and molecules.

Effect of heat on arrangement of particles

Investigate the effect of heat on particle motion during a change of states.

We have studied that matter is made up of particles. These particles are arranged in different ways in different kinds of matter and are moving in different ways. You have also studied in the previous class that matter changes its state on heating. Let us do an activity to investigate the effect of heating on particle motion.

Activity 5: Role play on heating matter

What I need:

- An outline of the classroom or a circle on the floor or school playground.
- A circle of approximately of 90 cm, in diameter.

What to do:

1) Mark a 90 cm diameter circle on the floor, with a rope, chalk or tape.
2) Have some students stand in the circle, close to each other, and jiggle.
3) After some time, stop them; ask them to look at where they are standing and how they are moving.
4) Explain that the circle is quite dense, with so many students standing close by in the circle, and that this is how particles in solid vibrate.
5) Now ask the students to imagine that the circle is heating up. Have the students move a little more and apart in a straight line. As they move, ask them to stop.
6) Ask them how close they are? Explain that they can move freely and the circle is becoming lighter and less dense as there are less people (particles) standing there, just as a liquid is usually less dense than a solid.
7) Now ask them to move around and out of the circle also. Explain that in particles gaseous state move freely. Stop them when only one child is left in the circle.
8) Explain that the circle is really light, just like a gas.

What I observed:

What movement did you observe in solid, liquid and gaseous state?

Did you observe how it becomes lighter as it changes state on heating?

What I concluded:

Teacher Note: Help the students imagine and observe the movement of particles in the three states of matter with the help of models. Discuss with the student while using various examples and activities to help students understand this important concept of particle arrangement and movement.

Did you conclude that matter changes from one state to another due to heating? As the matter **gains energy,** it changes state from **Solid to Liquid** and from **Liquid to Gases**. As energy increases from solid to liquid, the movement changes from vibration at a fixed point to movement at short distances, and then free movement in gases. Matter becomes lighter and less dense. Now, read and underline the key sentences about arrangement and movement of the particles in solid, liquids and gases.

After reading, complete the matrix on the next page.

Gases:

In gases, the particles or molecules are far apart and have low attractive forces between particles/molecules. Thus the particles in gaseous state move around everywhere and collide with one another. In the gaseous state, molecules move quickly and are free to move in any direction. Gases expand to fill their containers and have low density.

In gases, individual molecules are widely separated and can move around easily, gases can be compressed easily and they have an undefinite shape.

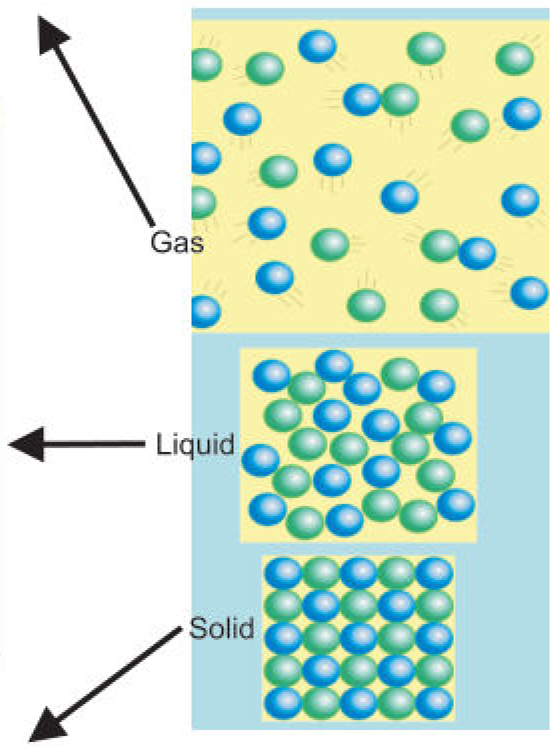
Liquids:

In Liquids, particles or molecules can move past one another, rotate and also change position; however, they remain relatively close to one another like solids.

As the temperature of a liquid is increased, the amount of movement of individual particles or molecules increases.

Thus, liquids can "flow" to take the shape of their container but they cannot be easily compressed because the molecules are already close together, constantly rotating, sliding and translating thus allowing no space to exist.

Thus, liquids have no shape, but a definite volume.



Solids:

In solids, the particles are close and have strong attractive forces between the individual particles or molecules. In solids, the individual particles or molecules are locked in a position and cannot move past one another.

The particles, atoms or molecules in solids remain in motion. However, that motion is limited to vibrational motion; individual particles/molecules stay fixed in a place and vibrate next to one another. As the temperature of a solid is increased, the amount of vibration increases, but the solid retains its shape.

In solids the particles are closely packed in place relative to one another thus solids have a high density.

Complete the matrix

States of matter	Arrangement of particles	Movement of particles	Characteristics
Gas			
Liquid			
Solid			

Processes involving change of states (melting, freezing, boiling, evaporation and condensation).

- Demonstrate and explain the processes that are involved in change of states.

What will happen to an ice cube on heating? As you have studied in the previous class, on heating the particles in the ice cubes will gain energy, move faster and change to liquid water. This is called melting. What will happen if the heating of the ice cube is continued? The ice cubes will convert to water. If heating is continued, the water will gain more energy, the temperature will rise and at a point, it will boil. What will happen if the water is kept in a freezer for a day? In freezer the water particles will release energy, move less and change to frozen ice.

During the process, more water will also evaporate and convert to a gaseous state of water, called the water vapour. If the water vapours are cooled, energy will be released and the gaseous state will convert to liquid water. This process is called condensation. These conversion in states is shown in the diagram below. Talk to your friend about it.

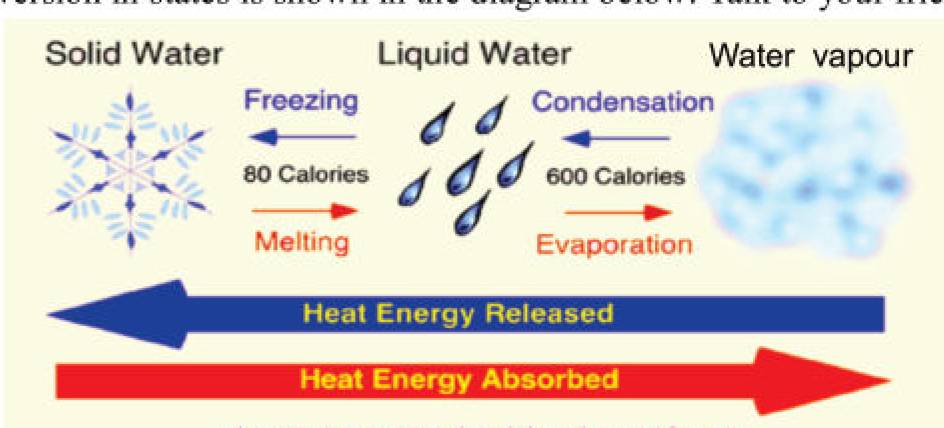


Figure 5.6: Processes involving change of states

Application of condensation and evaporation in nature (water cycle)

Activity 6: Describing the role of evaporation and condensation in water cycle and different form of moisture.

Describe the role of evaporation and condensation in the water cycle.

The story of water:

Water: "I am a water molecule, Do you know the places I travel during the water cycle and the states I live in?" Student: "No, could you tell me the places?"

Water: "I live in the oceans, rivers and lakes; When the sun shines on me, I get converted to water vapours and travel towards the sky"? Student: "What happens to you, water in the sky?"

Water: "As I rise up ward and travel higher in to cooler places, I get converted to water droplets and form clouds."

Students: " Ahaa! So that is how we get clouds after a very warm day. How do we get snow clouds"?

Water: "Yes, that is how I form clouds of water droplets, but for forming snow clouds, I have to travel from higher to colder region, so that I could cool down as clouds and form snow clouds?"

Students: "What happens after the snow and rain clouds are formed?"

Water: "Once these form into larger and heavier clouds, they fall back on hills and land as snow, hail and rain?" Student: "Yes, I have observed this rainfall from the large clouds and snowfall too. We also collect the rain water in tank on our roof. So is that all?" Water: "Next, as snow, I live on the mountain tops and as water I travel back into the oceans, lakes, and also under the ground as underground water."

Student: "Now I know, the water cycle, water you have an interesting cycle and you convert into many forms and also travel to several places. Thank you for telling me your story and how we get water." Observe the water cycle and stages below.

Teacher Note: Engage the students in a role play or a puppet show and have a dialogue on water cycle.

Heating and the states of water in water cycle

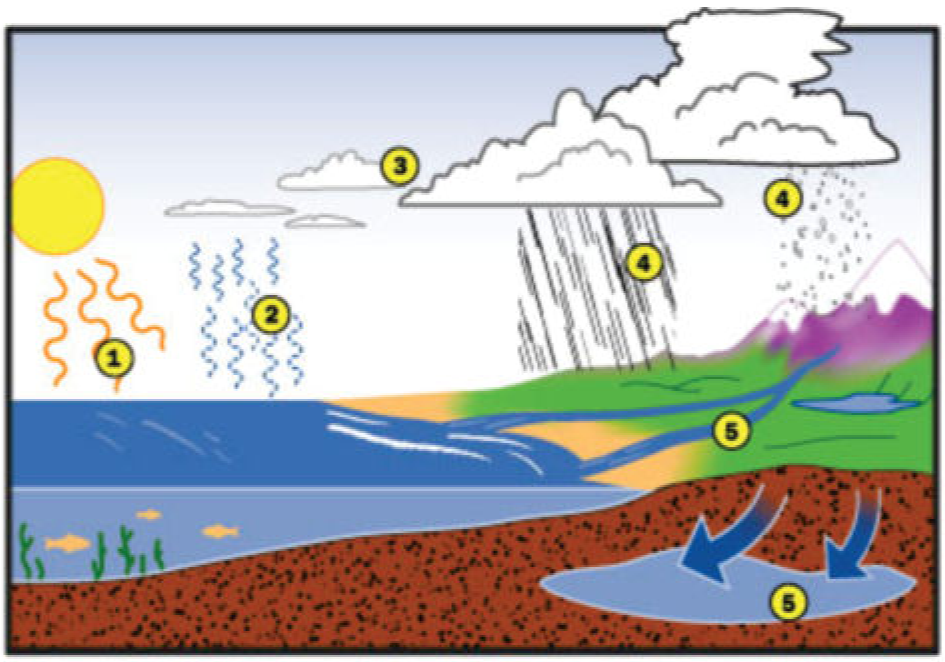


Figure 5.7:
The water cycle

1. The sun heats the ocean.
2. Ocean water evaporates and rises in to the air.
3. The water vapor cools and condenses to become droplets, which form clouds.
4. If enough water condenses, the drops become heavy enough to fall to the ground as rain and snow.
5. Some rain collects in groundwells. The rest flows through rivers back into the ocean.

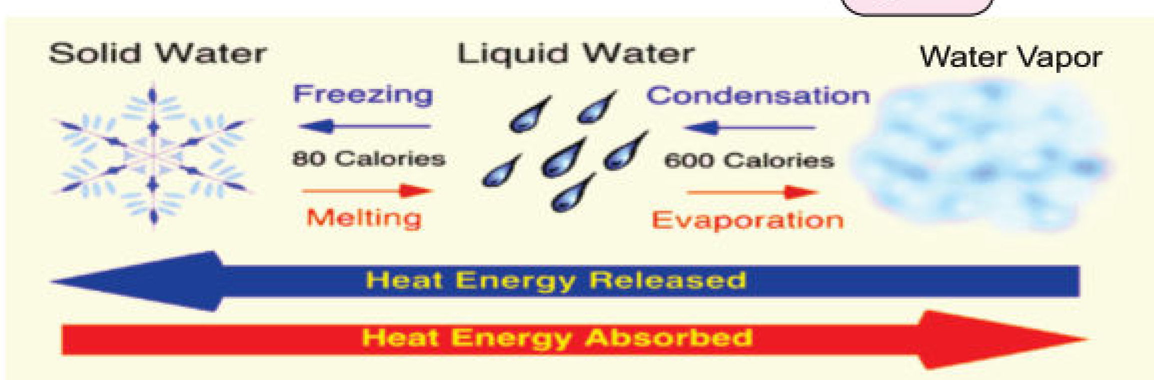
Do you know that the water cycle moves a lot of energy along with the water during the **heating** process, the conversion of **water** into **water vapour (gas)** and **cooling** process conversion of **water vapours** back into water **droplets (liquid)** and **snow (solid).** In this way the water cycle can transport huge amounts of **energy** into the **atmosphere**. The water cycle also **transports water** and naturally **purifies** water.

Identify and describe forms of moisture in the environment (e.g. dew, snow, fog, frost, and rain).

Do you know the forms of moisture in the Environment?
Dew: Drops of water that form on cool surfaces at night, due to condensation.
Snow: Water in the atmosphere frozen into crystals which falls as white flakes.
Fog: A thick cloud of water droplets on the earth's surface
Rain: Condensed water in the atmosphere of clouds that falls to the earth.

Summary

- Matter
  - Solid
    - Particle's arrangement: regular and closely packed
    - Vibrate at a fixed point
  - Liquid
    - Particle's arrangement: irregular and less closely packed
    - Move short distances, rotate and slide
  - Gas
    - Particles have no regular pattern, have large empty spaces
    - Move freely in any direction at long distances



Moisture in the environment

- Moisture in the environment
  - Dew
  - Fog
  - Frost
  - Rain
  - Snow

Review questions:

1. Circle T for True and F for False statements :

a. Particles in solids freely move around. T F

b. Particles in gas move around a fixed point. T F

c. Particles in liquids move short distances, rotate and slide. T F

d. Particles in liquid are regularly arranged and are closely packed. T F

e. Particles in gaseous state are irregular and have large spaces. T F

2. Give reasons to justify your answers:

A. Ice changes in to liquid water on heating.
B. Water vapour condenses to form liquid water.
C. Moisture exists in different forms.

Scientific problem solving:

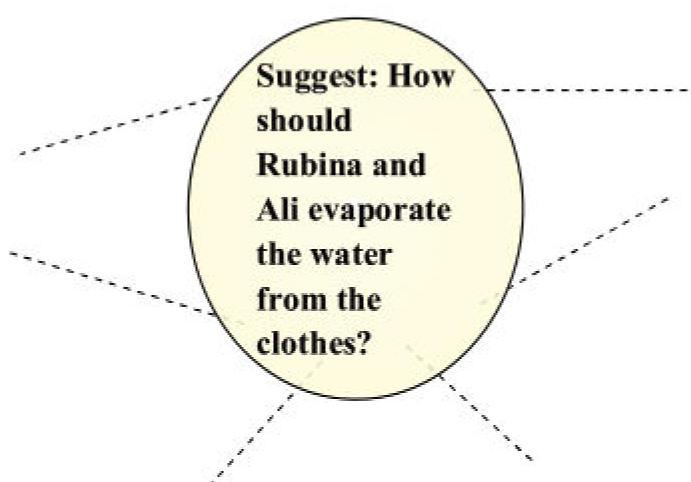
Story:

Mother was washing the school uniforms of Rubina and Ali. She wanted to dry the clothes before it gets dark. Mother was very worried how to dry the clothes quickly.

So Ali and Rubina set out to explore and dry the clothes.

How can you help Ali and Rubina in quickly drying the clothes?

Where should they hang the clothes?



Teacher Note: Guide the students to work in pairs or groups to brainstorm and record the steps and engage them in carrying out the investigation.

Activity 7: Evaporation at different places

What I need:

- three small saucers/plates
- water
- a spoon

What to do:

1. Your teacher will form groups/pairs or two students in a team.
2. Ask each member of a team to collect the materials.
3. Place a spoonful of water in each plate and mark the outline of water.
4. At the same time, place a plate A in a dark, covered and cold place; place plate B in open room ; place the third plate C in the sunny in a open air.
5. Observe the water after intervals of 30 minutes, 60 minutes and 1 hour 30 minutes and two hours.
6. Record your observation.
7. In which plate/place did the water evaporate fastest?
8. Share the results with each other.

What I observed:

What I conclude:

Teacher Note: Engage the students in the inquiry project, planning, performing, observing and recording in groups.